

Peter Morganelli

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EDUCATION

Tufts University

Bachelor of Science in Computer Science (BSCS)

Medford, MA

Expected: May 2027

- **Relevant Coursework:** Data Structures, Algorithms, Concurrent Systems, Natural Language Processing, Machine Structure and Assembly

Trinity College Dublin

Semester Abroad

Dublin, Ireland

January - May 2026

EXPERIENCE

Software Engineer Intern | Python, Django, Neo4j, Docker, GitLab

MITRE Headquarters

Bedford, MA

May 2026 - Present

- Working with applications that operate on **Controlled Unclassified Information (CUI)** for the **Air Force**, maintaining traceability between source requirements, system-model data, and **LLM/RAG**-generated findings within approved data-handling applications
- **Parallelized** requirements-checking analysis workloads to process independent requirement sets **concurrently**, reducing end-to-end analysis time by **77%** and improving responsiveness for large Excel imports
- Contributed to a **RAG** and **Django-based** requirements analysis platform that integrates with a **Neo4j** database
- Presented to an audience of **12+ senior engineers** about efficient AI workflows, explaining **agent orchestration** and **AI model prompting**

Data Structures Teaching Fellow | C++, GitLab, Gradescope

Tufts University

Medford, MA

September 2024 - Present

- Grading **100+** submissions per semester for code **correctness**, **style**, and **modularity**, while setting the grading call on ambiguous criteria for the rest of the TA staff
- Supporting **200+** students through weekly **office hours** and the course **Piazza** forum, answering questions and endorsing correct answers
- **Revising** course project specifications and lecture material alongside **senior course staff** and faculty

Software Developer | React, TypeScript, TailwindCSS, Figma

Tufts JumboCode

Medford, MA

October 2023 - May 2025

- Developed a **full-stack** inventory website for a local Somerville theater with a group of **10+ developers**, implementing **RESTful API** handlers and several **React** components for the application
- Designed and shipped an **image-upload carousel** with per-item metadata and filter tags from a **Figma** prototype

PROJECTS

[Peter's Pizzeria](#) | React, Vite, Vercel Serverless, Upstash Redis, GSAP, Claude

May - July 2026

- Built a **full-stack** ordering system for my student-run pizzeria using a React SPA frontend with a live cart, per-item customization, and order tracking, backed by **Vercel serverless functions** with **Redis-based** persistence
- Designed a **real-time** kitchen dashboard with **polling-based** live updates, order aggregation for prep, and **HMAC-authenticated** admin routes
- Hardened the **backend** with **server-side** price validation, **rate limiting**, store-hour enforcement, and inventory management to prevent client-side tampering and abuse

FoodWaste Tracker | React Native + TypeScript, SpringBoot + Java, Firebase, PostgreSQL

January - May 2026

- **Integrated** end-to-end safe user **authentication** using **Firebase** service accounts and API keys
- **Developed** an **iOS** and **Android** compatible **mobile application** that aligned with a **UN Sustainability Development** goal
- Worked with a team of **six** developers for study abroad at **Trinity College Dublin** in Dublin, Ireland for the **Group Design Project** module

[Programming Language Interpreter](#) | Elm, JavaScript, HTML, CSS

May - August 2025

- **Engineered** a **zero-setup**, **web-based IDE** from **scratch**, integrating several interpreters (Impcore, nano-ML, Molecule, uScheme, uSmalltalk, etc.) via **Elm ports** and **JavaScript** bridges to enable **live program execution** in the browser
- Designed a modular interpreter **API** for new languages, suited for **200+** students to complete their homework
- Built under the mentorship of Professor Emiertus **Dr. Norman Ramsey**, author of the course textbook

SKILLS

Programming Languages: C++, C, C#, Python, JavaScript, TypeScript, x86 Assembly

Frameworks: React, React Native, Next.js, Django, Three.js, Unity, AGILE/Scrum

Infrastructure & Tools: Docker, AWS (S3), Vercel, PostgreSQL, Neo4j, Redis, Firebase, GitHub, GitLab, Linux